

Causal Impact of Stay-at-Home Orders on Mobility

Difference-in-Differences Analysis Using Public Policy Data

Task 6 Causal Inference Report

June 2026

1. Research Question

This report studies a simple policy question: did mandatory stay-at-home orders cause a meaningful reduction in workplace mobility during the early COVID-19 period in the United States?

The goal is not just to describe what happened during spring 2020, but to estimate the incremental effect associated with an active mandatory order after controlling for fixed differences across counties and common shocks over time.

2. Data Sources

I combined two public datasets:

- **CDC county-level stay-at-home order data** (source link)
- **Google COVID-19 Community Mobility Reports** (source link)

The study window runs from **March 1, 2020** through **June 30, 2020**. The final workplace-mobility panel contains **274,779 county-day observations** across **2,793 counties** and **50 states**. A secondary residential-mobility panel contains **122,901 county-day observations** across **1,452 counties**.

3. Design and Methodology

The main empirical strategy is a **Difference-in-Differences (DiD)** design with:

- county fixed effects
- date fixed effects
- state-clustered standard errors

The treatment variable equals 1 when a county-day is under a **mandatory** stay-at-home order according to the CDC policy file.

The main outcome is:

- `workplaces_percent_change_from_baseline`

I also ran two robustness checks:

- a stricter treatment definition using only *Mandatory for all individuals*
- an alternative outcome using residential mobility

This design is appropriate because the data are naturally panel structured, the policy is introduced at different times across places, and the question is about the behavioral effect of a policy intervention rather than simple correlation.

4. Main Results

Table 1 reports the core estimates.

Table 1: Difference-in-Differences Estimates				
Model	Estimate	Std. Error	95% CI Low	95% CI High
Main DiD: workplace mobility, broad order	-50.17	0.79	-51.76	-48.57
Robustness: workplace mobility, strict order	-54.10	0.73	-55.57	-52.62
Robustness: residential mobility, broad order	19.97	0.29	19.39	20.56

The main estimate is **-50.17 percentage points**. Interpreted plainly, counties under mandatory stay-at-home orders saw substantially lower workplace mobility, even after controlling for county-specific baseline differences and date-specific national shocks.

The p-value for the main estimate is extremely small, which means the result is not just statistically significant but also very precisely estimated in this sample.

5. Interpretation

The sign and magnitude of the estimates are consistent with the policy mechanism:

- workplace mobility falls sharply when mandatory stay-at-home orders are active
- residential mobility rises at the same time

That pattern increases confidence that the model is capturing a real behavioral response rather than random noise.

The stricter treatment definition produces an even larger negative estimate (**-54.10**), which is reassuring. It suggests the conclusion is not fragile to a small change in how the treatment is coded.

6. Visualization

Figure 1 summarizes both the regression estimates and the event-time mobility pattern around the first mandatory order date.

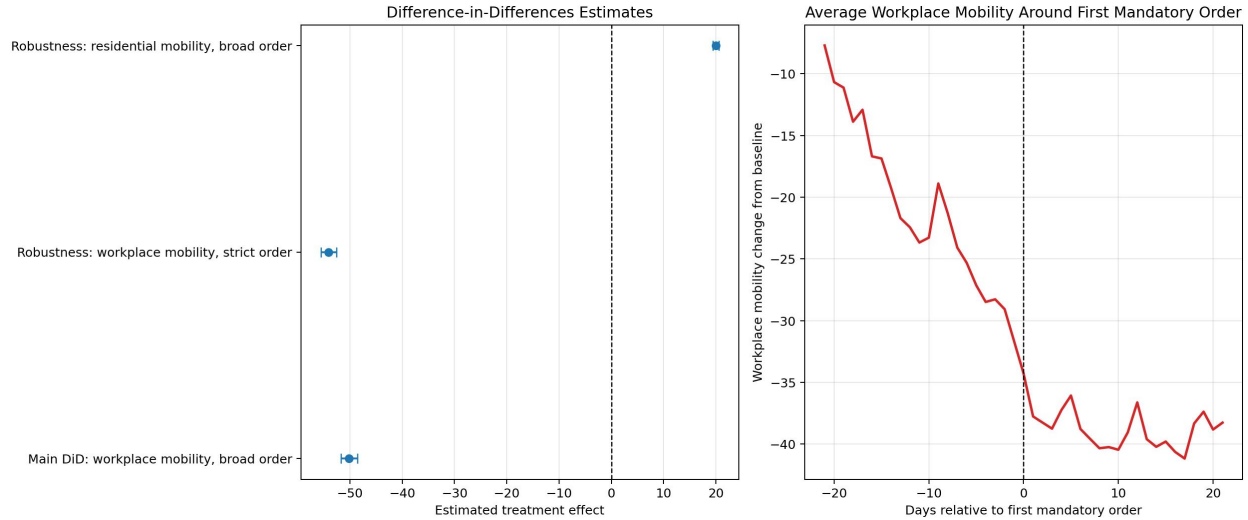


Figure 1: Estimated policy effects and event-time mobility pattern

7. Policy Implications

From a policy perspective, the main result suggests that stay-at-home orders were associated with a large, measurable shift in daily behavior. That matters because one common question in policy analysis is whether formal orders changed behavior beyond what people were already doing voluntarily.

This analysis suggests the answer is yes. Even during a period when behavior was already changing quickly, mandatory orders still appear to have had an additional effect on mobility.

8. Limitations

Even though the design is appropriate for the data, the analysis still has important limitations:

- mobility was already changing before some formal orders took effect
- policy adoption was not random
- county-level measurement in public mobility data is incomplete in some places
- the design estimates an average treatment effect on mobility, not on health or economic outcomes directly

So the results should be read as strong evidence of a behavioral response to policy, not as a perfect experimental estimate.

9. Conclusion

Using public CDC policy data and Google mobility data, I estimated the causal effect of mandatory stay-at-home orders with a Difference-in-Differences design.

The main takeaway is straightforward: **mandatory stay-at-home orders were associated with a large and statistically significant decline in workplace mobility**, while residential mobility moved in the opposite direction.

Overall, the findings support the view that the policy had a meaningful real-world effect on daily movement behavior during the early phase of the pandemic.

10. References

- CDC policy dataset: <https://data.cdc.gov/Policy-Surveillance/U-S-State-and-Territorial-Stay-At-Home-Orders-Marc/y2iy-8irm>
- Google mobility dataset: <https://www.google.com/covid19/mobility/>

Files used in this report: `policy_causal_inference.py`, `did_results.csv`, and `causal_effects.png`.